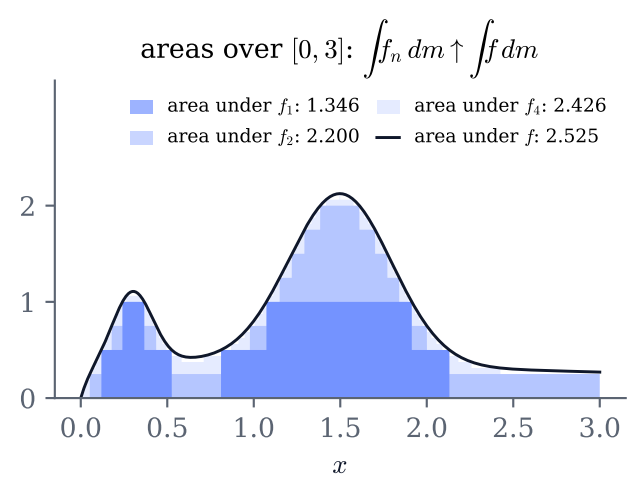
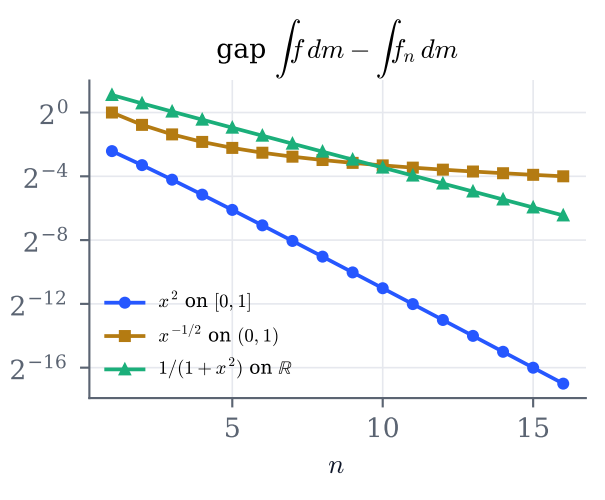

areas over $[0, 3]$: $\int f_n\,dm \uparrow \int f\,dm$
area under $f_1$: 1.346
area under $f_4$: 2.426
area under $f_2$: 2.200
area under $f$: 2.525
0
1
2
0.0
0.5
1.0
1.5
2.0
2.5
3.0
$x$


gap $\int f\,dm - \int f_n\,dm$
$2^{0}$
$2^{-4}$
$2^{-8}$
$2^{-12}$
$2^{-16}$
$x^2$ on $[0, 1]$
$x^{-1/2}$ on $(0, 1)$
$1/(1+x^2)$ on $\mathbb{R}$
5
10
15
$n$